

mb-pc40s-2016-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

mili

A pizza with a diameter of 15 inches is cut into equal slices, each with a central angle of 36° .

Determine the length of the crust on the outer edge of one slice of pizza.

Solution

$$\begin{aligned}\frac{36^\circ}{180^\circ} &= \frac{\theta}{\pi} && 1 \text{ mark for conversion} \\ \theta &= \frac{\pi}{5} \\ s &= \theta r \\ s &= \left(\frac{\pi}{5}\right)\left(\frac{15}{2}\right) && 1 \text{ mark for substitution} \\ s &= \frac{3\pi}{2} \text{ inches} && 2 \text{ marks} \\ \text{or} \\ s &= 4.712 \text{ inches}\end{aligned}$$

Question 2

m

There are 9 girls and 7 boys in a math class from which a committee of 5 is to be chosen.

a) How many different committees of 5 can be formed if one of the boys, William, must be on the committee?

b) How many different committees of 5 can be formed if there must be 2 girls and 3 boys on the committee?

Solution

a)

$${}_1C_1 \cdot {}_{15}C_4 = 1365$$

b)

$${}_9C_2 \cdot {}_7C_3 = 1260$$

$\frac{1}{2}$ mark for ${}_9C_2$

$\frac{1}{2}$ mark for ${}_7C_3$

1 mark for the product of combinations

2 marks

Note(s):

- ${}_1C_1$ does not need to be shown

Question 3

Solve the following equation over the interval $[0, 2\pi]$:

$$\sin^2 \theta + 6 \sin \theta - 2 = 0$$

Solution

$$\sin \theta = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(-2)}}{2(1)}$$

$$\sin \theta = \frac{-6 \pm \sqrt{36+8}}{2}$$

$$\sin \theta = \frac{-6 \pm \sqrt{44}}{2}$$

$$\sin \theta = 0.316\ 624...$$

$$\sin \theta = -6.316624...$$

$$\theta_r = 0.322169$$

$$\theta = 0.322$$

no solution

$$\theta = 2.819$$

1 mark for solving for $\sin \theta$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value, 1 mark for indicating no solution)

3 marks

Question 4

Solve:

$$6(5)^{3x+2} = 9^{2-x}$$

Solution

$\log[6(5)^{3x+2}] = \log 9^{2-x}$	$\frac{1}{2}$ mark for applying logarithms
$\log 6 + \log 5^{3x+2} = \log 9^{2-x}$	1 mark for product law
$\log 6 + (3x + 2) \log 5 = (2 - x) \log 9$	1 mark for power law
$\log 6 + 3x \log 5 + 2 \log 5 = 2 \log 9 - x \log 9$	
$x(3 \log 5 + \log 9) = 2 \log 9 - 2 \log 5 - \log 6$	$\frac{1}{2}$ mark for collecting terms with x
$x = \frac{2 \log 9 - 2 \log 5 - \log 6}{3 \log 5 + \log 9}$	$\frac{1}{2}$ mark for solving for x
$x = -0.087707$	$\frac{1}{2}$ mark for evaluating quotient of logarithms
$= -0.088$	4 marks

Question 5

Solve $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ where $\theta \in \mathbb{R}$.

Solution

$$2 \sin \theta - 1 = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta_r = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\sin \theta + 1 = 0$$

$$\theta_r = \frac{3\pi}{2}$$

1 mark for solving for $\sin \theta$

2 marks for solving for θ (1 mark for each branch)

1 mark for general solution

$$\theta_r = 30^\circ$$

$$\theta = 30^\circ + 360^\circ k, k \in \mathbb{Z}$$

$$\theta = 150^\circ + 360^\circ k, k \in \mathbb{Z}$$

$$\theta_r = 270^\circ$$

$$\theta = 270^\circ + 360^\circ k, \quad k \in \mathbb{Z}$$

4 marks

Question 6

The roots of the polynomial equation $3(x - 2)(x + 1)^2 = 0$ are $x = 2$ and $x = -1$.

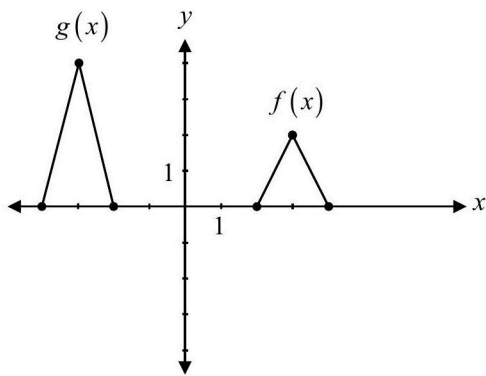
Explain what these roots represent on the graph of $p(x) = 3(x - 2)(x + 1)^2$.

Solution

They are the x -intercepts of the graph of $p(x)$.

Question 7

Determine an equation for $g(x)$ as a transformation of $f(x)$.



$g(x) =$

Solution

$g(x) = 2f(x + 6)$	1 mark for vertical stretch 1 mark for horizontal translation
	2 marks
or	
$g(x) = 2f(-x)$	1 mark for vertical stretch 1 mark for horizontal reflection
	2 marks

Question 8

A student must determine the factors of $5x^4 - 2x^3 + 4x - 1$. He used 5, -2 , 4, and -1 as the coefficients of the polynomial when using synthetic division.

Explain the student's error.

Solution

The student did not write the coefficient of 0 for the x^2 term.

Question 9

Describe the transformations of $y = f(x)$ when asked to sketch the graph of $y = -f(x - 4)$.

Solution

$f(x)$ is reflected over the x -axis and translated 4 units to the right.

1 mark for vertical reflection

1 mark for horizontal translation

2 marks

Question 10

Prove the identity below for all permissible values of θ :

$$\sin \theta + \frac{\cos \theta}{\tan \theta} = \frac{1}{\cos \theta \tan \theta}$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\sin \theta + \frac{\cos \theta}{\tan \theta}$	$\frac{1}{\cos \theta \tan \theta}$
$\sin \theta + \frac{\cos \theta}{\frac{\sin \theta}{\cos \theta}}$	$\frac{1}{\cos \theta \frac{\sin \theta}{\cos \theta}}$
	$\frac{1}{\sin \theta}$

1 mark for correct substitution of identities

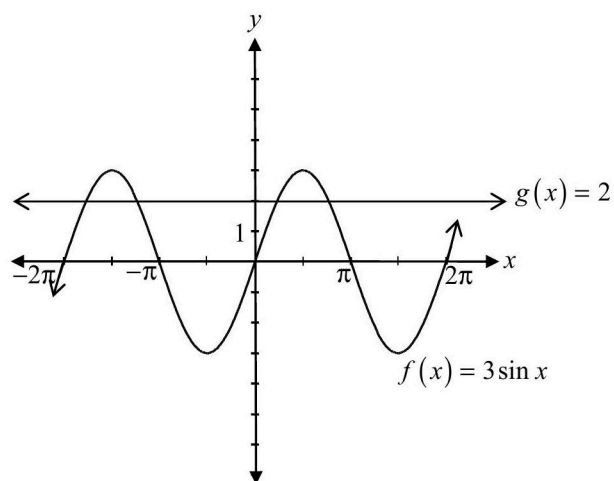
1 mark for algebraic strategies

1 mark for logical process to prove an identity

3 marks

Question 11

Describe how to use the graphs of $f(x) = 3 \sin x$ and $g(x) = 2$ to solve the equation $3 \sin x = 2$.



Solution

The solution will be the x -values where the two graphs intersect.

Question 12

A hockey arena has 5 doors.

Determine the number of ways that you can enter through one door but exit through a different door.

Solution

$$5 \cdot 4 = 20 \text{ ways}$$

Question 13

Given that $(x + 3)$ is one of the factors, express $2x^3 + 7x^2 + 2x - 3$ as a product of factors.

Solution

-3		7	2	-3
	2	1	-1	0

$(x + 3)(2x^2 + x - 1)$

or

$(x + 3)(2x - 1)(x + 1)$

$\frac{1}{2}$ mark for $x = -3$

1 mark for synthetic division (or for any other equivalent strategy)

$\frac{1}{2}$ mark for consistent product of factors

2 marks

Question 14

Identify the maximum number of x -intercepts for a polynomial function of degree 3 .

A.

1

B.

2

C.

3

D.

4

Solution

Correct answer: C

3

Question 15

The graph of $y = f(x)$ contains the point (a, b) . The graph of $g(x)$ is a transformation of the graph of $f(x)$ and contains the point $(3a, b)$.

Identify the function that represents $g(x)$.

A.

$$g(x) = f(3x)$$

B.

$$g(x) = 3f(x)$$

C.

$$g(x) = f\left(\frac{x}{3}\right)$$

D.

$$g(x) = \frac{1}{3}f(x)$$

Solution

Correct answer: C

$$g(x) = f\left(\frac{x}{3}\right)$$

Question 16

The angle 2.95 radians, in standard position, terminates in quadrant:

A.

I

B.

II

C.

III

D.

IV

Solution

Correct answer: B

II

Question 17

Evaluate:

$$2 \sin \frac{\pi}{8} \cos \frac{\pi}{8}$$

A.

$$\frac{1}{2}$$

B.

$$\frac{\sqrt{2}}{2}$$

C.

$$1$$

D.

$$\sqrt{2}$$

Solution

Correct answer: B

$$\frac{\sqrt{2}}{2}$$

Question 18

Identify which of the following represents the 5th term in the expansion of $(4x^2 - 2y^3)^{15}$.

A.

$${}_{15}C_5(4x^2)^{10}(-2y^3)^5$$

B.

$${}_{15}C_5(4x^2)^{11}(-2y^3)^4$$

C.

$${}_{15}C_4(4x^2)^{10}(-2y^3)^5$$

D.

$${}_{15}C_4(4x^2)^{11}(-2y^3)^4$$

Solution

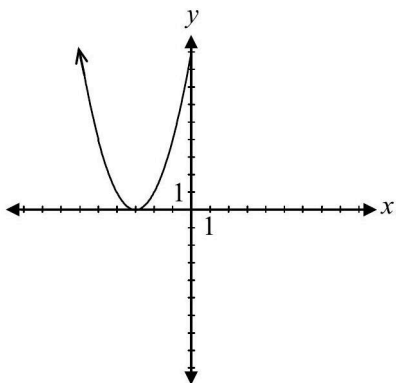
Correct answer: D

$${}_{15}C_4(4x^2)^{11}(-2y^3)^4$$

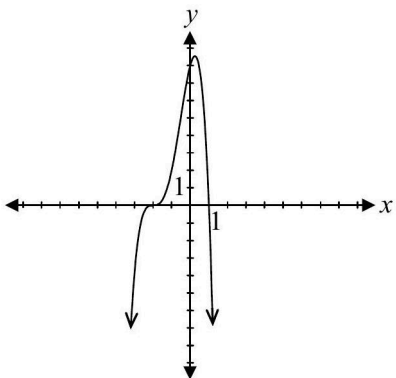
Question 19

Identify which of the following graphs of polynomial functions has a zero with a multiplicity of 3.

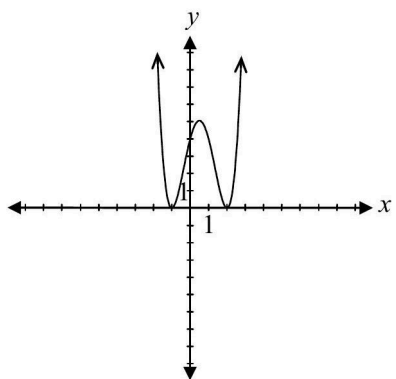
A.



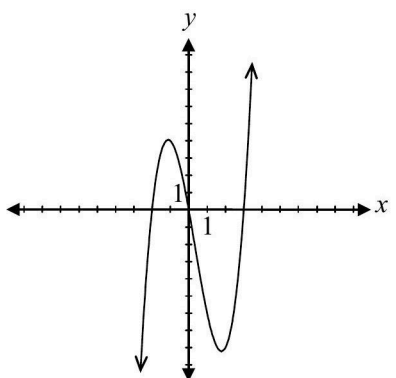
B.



C.

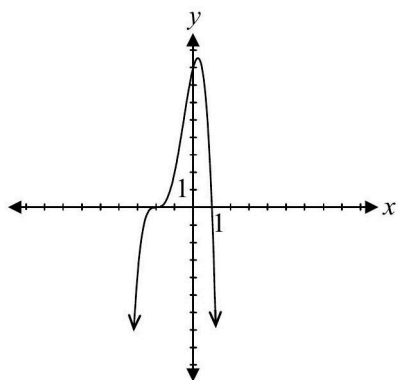


D.



Solution

Correct answer: B



Question 20

A non-permissible value of x for the function $f(x) = \frac{1}{\cos x + 1}$ is:

A.

-1

B.

0

C.

π

D.

$\frac{3\pi}{2}$

Solution

Correct answer: C

π

Question 21

Identify which of the following statements is true for the rational function $f(x) = \frac{4(x-1)(x-2)}{(x-1)(x+3)}$.

A.

The equation of the horizontal asymptote is $y = 4$.

B.

The equation of the vertical asymptote is $x = 1$.

C.

The y -intercept is 0 .

D.

There is a point of discontinuity (hole) when $x = 2$.

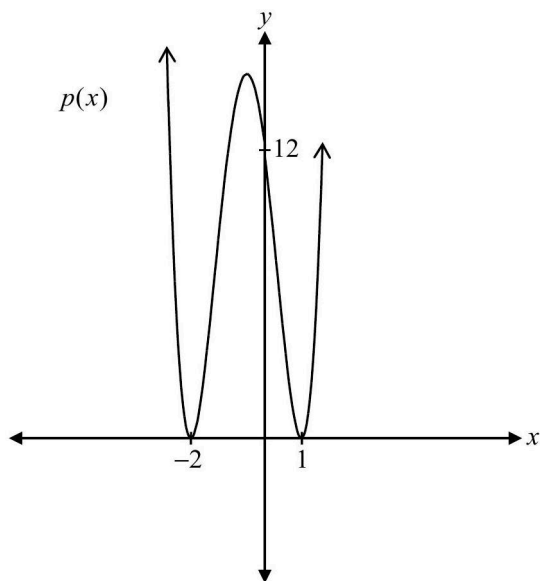
Solution

Correct answer: A

The equation of the horizontal asymptote is $y = 4$.

Question 22

Determine the equation of the polynomial function, $p(x)$, represented by the graph.



$$p(x) =$$

Solution

$$p(x) = 3(x + 2)^2(x - 1)^2$$

1 mark for factors

1 mark for multiplicity of 2 ($\frac{1}{2}$ mark for each)

1 mark for correct value of a (consistent with factors and multiplicity)

3 marks

Question 23

Evaluate:

$$\log_4 2$$

Solution

$$\frac{1}{2}$$

Question 24

Evaluate:

$$\left(\cos \frac{11\pi}{3}\right)\left(\csc \frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right)(-2)$$

1 mark for $\cos \frac{11\pi}{3}$ ($\frac{1}{2}$ mark for the quadrant, $\frac{1}{2}$ mark for the value)

1 mark for $\csc \frac{11\pi}{6}$ ($\frac{1}{2}$ mark for the quadrant, $\frac{1}{2}$ mark for the value)

-1

2 marks

Question 25

Estimate the value of $\log_2 5$.

Justify your answer.

Solution

$$\left. \begin{array}{l} \log_2 4 = 2 \\ \log_2 8 = 3 \end{array} \right\} \frac{1}{2} \text{ mark for justification}$$

1 mark

Question 26

If θ terminates in quadrant III and $\cos \theta = -\frac{6}{7}$, determine the exact value of $\tan \theta$.

Solution

$$\cos \theta = \frac{x}{r}$$

$$x^2 + y^2 = r^2$$

$$y^2 = (7)^2 - (-6)^2 \quad 1/2 \text{ mark for substitution of } x = -6 \text{ and } r = 7$$

$$y^2 = 13$$

$$y = \pm\sqrt{13} \quad 1/2 \text{ mark for solving for } y$$

$$\tan \theta = \frac{\sqrt{13}}{6} \quad 1 \text{ mark for the value of } \tan \theta \left(\frac{1}{2} \text{ mark for the quadrant, } \frac{1}{2} \text{ mark for the value} \right)$$

Question 27

Given $f(x) = x^2 + x - 4$ and $g(x) = \sqrt{x + 5}$, Taz was asked to find $f(g(x))$.

Taz's solution:

$$\begin{aligned}f(g(x)) &= (\sqrt{x + 5})^2 + x - 4 \\&= x + 5 + x - 4 \\&= 2x + 1, x \geq -5\end{aligned}$$

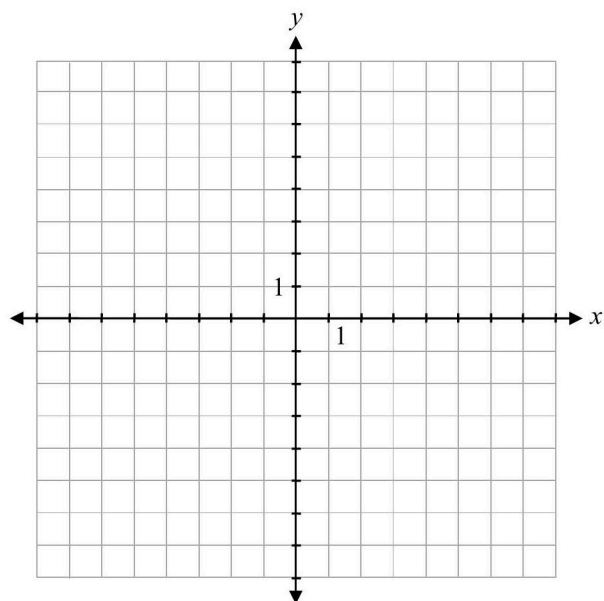
Describe the error in Taz's solution.

Solution

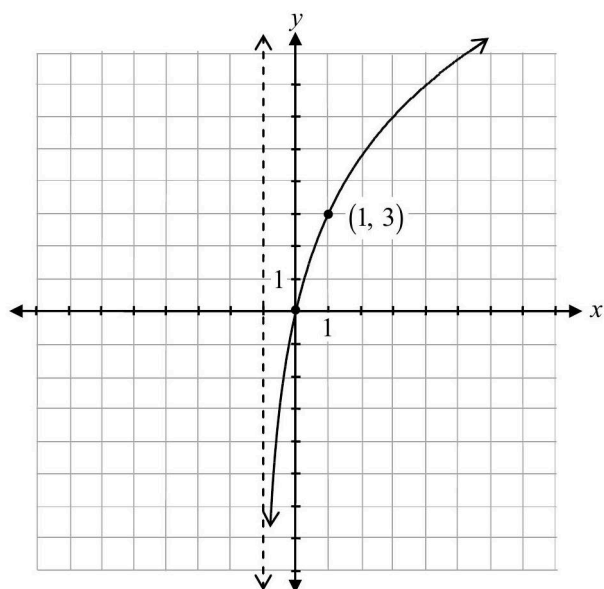
Taz must substitute $g(x)$ in both terms containing x in $f(x)$ and then simplify.

Question 28

Sketch the graph of the function $f(x) = 3 \log_2(x + 1)$.



Solution



1 mark for increasing logarithmic function

1 mark for vertical stretch

1 mark for asymptotic behaviour at $x = -1$

3 marks

Question 29

Write an equation of a rational function that would not have any vertical asymptotes.

Solution

Various equations, such as the following, are possible:

$$y = \frac{(x-2)(x+1)}{(x-2)}$$

or

$$y = \frac{4}{x^2 + 4}$$

Question 30

Determine the exact value of $\tan 75^\circ$.

Solution

$$\begin{aligned}\tan 75^\circ &= \tan(30^\circ + 45^\circ) \\&= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ} \\&= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}(1)} \\&= \frac{\frac{1+\sqrt{3}}{\sqrt{3}}}{\frac{\sqrt{3}-1}{\sqrt{3}}} \\&= \frac{1 + \sqrt{3}}{\sqrt{3} - 1}\end{aligned}$$

or

$$= \frac{\sqrt{3} + 3}{3 - \sqrt{3}}$$

1 mark for combination

1 mark for exact values ($\frac{1}{2}$ mark for each)

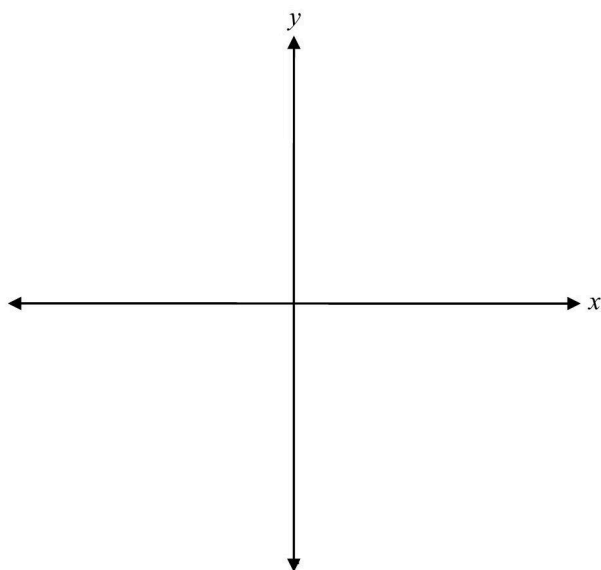
2 marks

- Other combinations are possible.

Question 31

Sketch the graph of the following function:

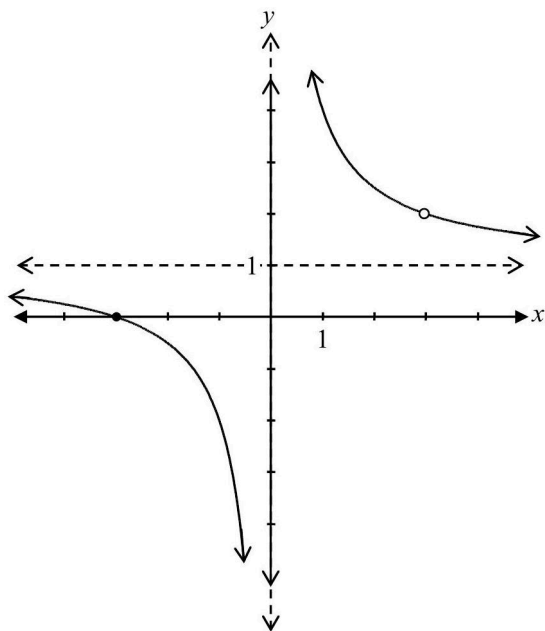
$$f(x) = \frac{(x+3)(x-3)}{x(x-3)}$$



Solution

$$\begin{aligned} f(x) &= \frac{(x+3)(x-3)}{x(x-3)} \\ &= \frac{x+3}{x}, x \neq 3 \end{aligned}$$

$\therefore$ there is a point of discontinuity (hole) at $(3, 2)$



1 mark for asymptotic behaviour at $y = 1$

1 mark for asymptotic behaviour at $x = 0$

1 mark for point of discontinuity (hole) at $(3, 2)$

($\frac{1}{2}$ mark for $x = 3$, $\frac{1}{2}$ mark for $y = 2$)

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

4 marks

Question 32

In the binomial expansion of $\left(\frac{1}{x^3} - 2x^2\right)^9$, determine which term contains x^3 .

Solution

Method 1

$$x^3 = (x^{-3})^{9-k} (x^2)^k$$

$$x^3 = x^{-27+3k+2k}$$

$$3 = -27 + 3k + 2k$$

$$30 = 5k$$

$$6 = k$$

$\therefore$ term 7 would contain x^3

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for solving for k

1 mark for identifying the 7th term

(or consistent term with the value of k)

2 marks

Method 2

$$\left(\frac{1}{x^3}\right)^9, \left(\frac{1}{x^3}\right)^8 (x^2), \left(\frac{1}{x^3}\right)^7 (x^2)^2$$
$$x^{-27}, x^{-22}, x^{-17}$$

$\therefore$ term 7 would contain x^3

1 mark for determining the pattern

1 mark for identifying the 7th term

(or consistent term with the pattern)

2 marks

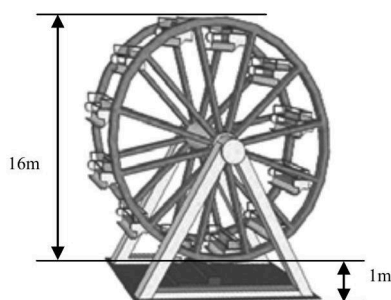
Question 33

José and Dana get on a Ferris wheel, which is 1 metre off the ground. The diameter of the Ferris wheel is 16 metres. Their ride lasts for 4 minutes, in which time the Ferris wheel makes one revolution.

Determine the values of A, B, C, and D , if the sinusoidal function that models the situation is $h(t) = A \cos[B(t - C)] + D$, where h is the height at which José and Dana are located on the Ferris wheel, from the ground, in metres, and t is the time, in minutes.

A =

B =



C =

D =

Solution

$$A = \frac{8}{\frac{\pi}{2}}$$

$$B = \frac{2}{9}$$

$$C = \frac{2}{D}$$

$$D = 9$$

$$A = \frac{-8}{B=}$$

1 mark for A

$$C = \frac{\frac{\pi}{2}}{0}$$

1 mark for B

or

$$D = \frac{9}{1 \text{ mark for } C}$$

1 mark for D

4 marks

Note(s)

- Other answers are possible.

Question 34

Solve algebraically:

$${}_nP_3 = 4!(n-1)$$

Solution

$$\frac{n!}{(n-3)!} = 4!(n-1)$$
$$\frac{n(n-1)(n-2)(n-3)!}{(n-3)!} = 4!(n-1)$$

$$n(n-2) = 24$$

$$n^2 - 2n - 24 = 0$$

$$(n-6)(n+4) = 0$$

$$n = 6 \text{ or } n = -4$$

$\frac{1}{2}$ mark for substitution

1 mark for factorial expansion

$\frac{1}{2}$ mark for simplification of factorials

$\frac{1}{2}$ mark for rejecting extraneous root

$\frac{1}{2}$ mark for the value of n

3 marks

Question 35

Given $f(x) = \frac{2}{x-1}$, determine the equation of the inverse, $f^{-1}(x)$.

Solution

Method 1

let $y = f(x)$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

$$y-1 = \frac{2}{x}$$

$$y = \frac{2}{x} + 1$$

$$f^{-1}(x) = \frac{2}{x} + 1$$

1 mark for switching x and y values

$\frac{1}{2}$ mark for solving for y

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

Method 2

let $y = f(x)$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

$$x(y-1) = 2$$

$$xy - x = 2$$

$$xy = 2 + x$$

$$y = \frac{2+x}{x}$$

$$f^{-1}(x) = \frac{2+x}{x}$$

2 marks

1 mark for switching x and y values

$\frac{1}{2}$ mark for solving for y

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Question 36

Solve:

$$4 \log_3 2 - \frac{1}{3} \log_3 8 = \log_3 a$$

Solution

Method 1

$$\log_3 2^4 - \log_3 8^{\frac{1}{3}} = \log_3 a$$

$$\log_3 16 - \log_3 2 = \log_3 a$$

$$\log_3 \left(\frac{16}{2} \right) = \log_3 a$$

$$\log_3 8 = \log_3 a$$

$$a = 8$$

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

1 mark for equating arguments

3 marks

Method 2

$$\log_3 2^4 - \log_3 8^{\frac{1}{3}} - \log_3 a = 0$$

$$\log_3 16 - \log_3 2 - \log_3 a = 0$$

$$\log_3 \left(\frac{16}{2a} \right) = 0$$

$$\log_3 \left(\frac{8}{a} \right) = 0$$

$$3^0 = \frac{8}{a}$$

$$1 = \frac{8}{a}$$

$$a = 8$$

1 mark for power law ($\frac{1}{2}$ mark for each)

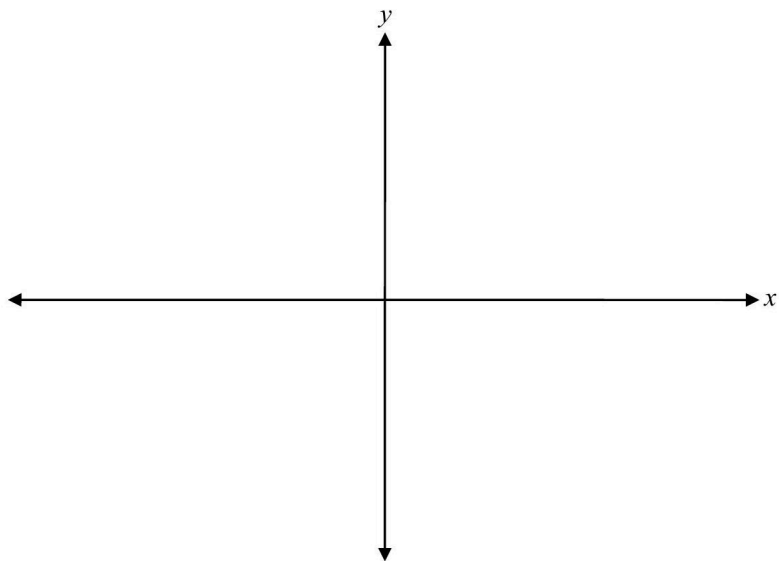
1 mark for quotient law

1 mark for exponential form

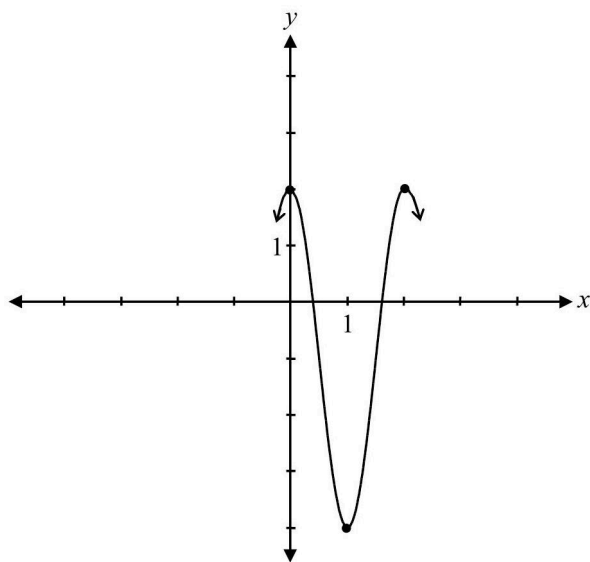
3 marks

Question 37

Sketch the graph of at least one period of the function $y = 3 \cos(\pi x) - 1$.



Solution



$$\begin{aligned}\text{period} &= \frac{2\pi}{\pi} \\ &= 2\end{aligned}$$

1 mark for amplitude

1 mark for period

1 mark for vertical translation

3 marks

Question 38

Using the laws of logarithms, fully expand the expression:

$$\log_a \left(\frac{x^3}{y\sqrt{z}} \right)$$

Solution

$$\begin{aligned}\log_a \left(\frac{x^3}{y\sqrt{z}} \right) &= \log_a x^3 - (\log_a y + \log_a \sqrt{z}) \\ &= 3 \log_a x - \left(\log_a y + \frac{1}{2} \log_a z \right) \\ &= 3 \log_a x - \log_a y - \frac{1}{2} \log_a z\end{aligned}$$

1 mark for quotient law

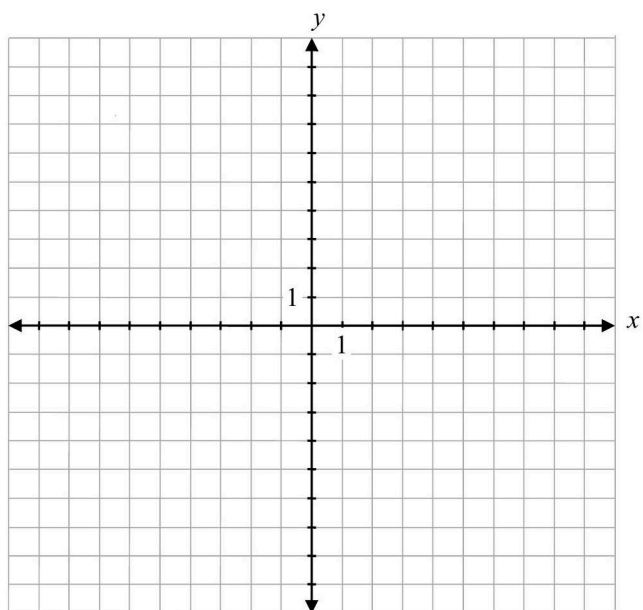
1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

3 marks

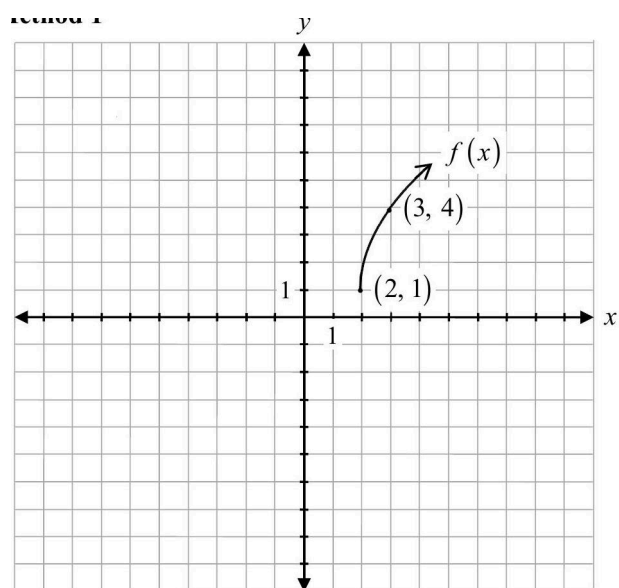
Question 39

Sketch the graph of $f(x) = 3\sqrt{x-2} + 1$.



Solution

Method 1



1 mark for horizontal translation

1 mark for vertical translation

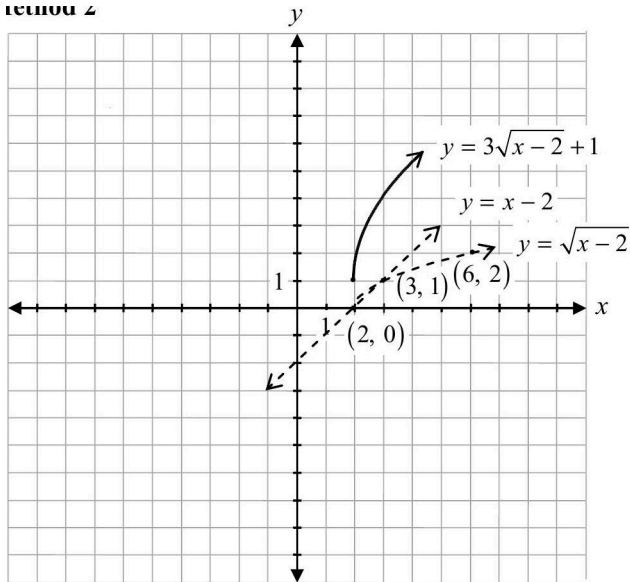
1 mark for shape of a radical function

1 mark for vertical stretch

4 marks

Method 2

Method 2



1 mark for invariant points where $y = 0$

and $y = 1$ ($\frac{1}{2}$ mark for each point)

1 mark for domain $[2, \infty)$

$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of the invariant points

1 mark for applying transformations

($\frac{1}{2}$ mark for vertical stretch, $\frac{1}{2}$ mark for vertical translation)

4 marks

Question 40

a) Determine the domain of the graph of the function $f(x) = \sqrt{x^2 - 4}$.

b) Explain why the domain of $f(x) = \sqrt{x^2 - 4}$ is restricted.

Solution

a)

$$\{x \mid x \leq -2 \cup x \geq 2\}$$

or

$$D : (-\infty, -2] \cup [2, \infty)$$

1 mark for domain ($\frac{1}{2}$ mark for $x \leq -2$, $\frac{1}{2}$ mark

for $x \geq 2$)

1 mark

b) The domain is restricted because you cannot take

the square root of a negative number.

Question 41

a) 1 mark b) 1 mark $\frac{135}{136}$

Given the point $(-12, -18)$ on the graph of $f(x)$, determine the new points after the following transformations of $f(x)$.

a) $\frac{1}{f(x)}$

b) $f(-x) + 10$

Solution

a) $(-12, \frac{-1}{18})$

b) $(12, -8)$

1 mark ($\frac{1}{2}$ mark for x -value, $\frac{1}{2}$ mark for y -value)

1 mark

Question 42

Explain why there is no solution for the equation $\csc \theta = -\frac{1}{2}$.

Solution

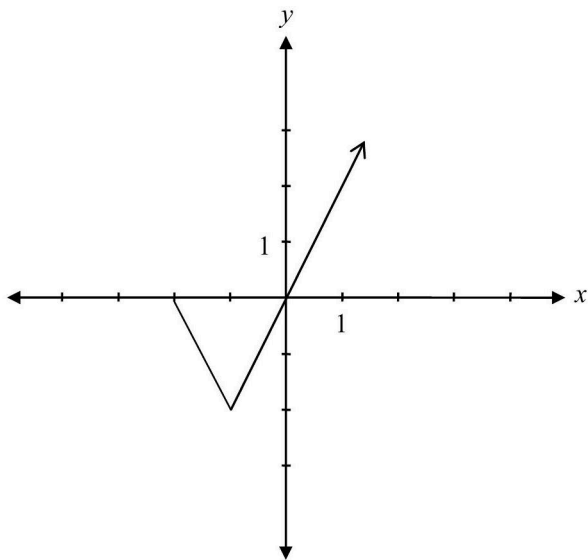
The value of $\csc \theta$ cannot be between -1 and 1 .

or

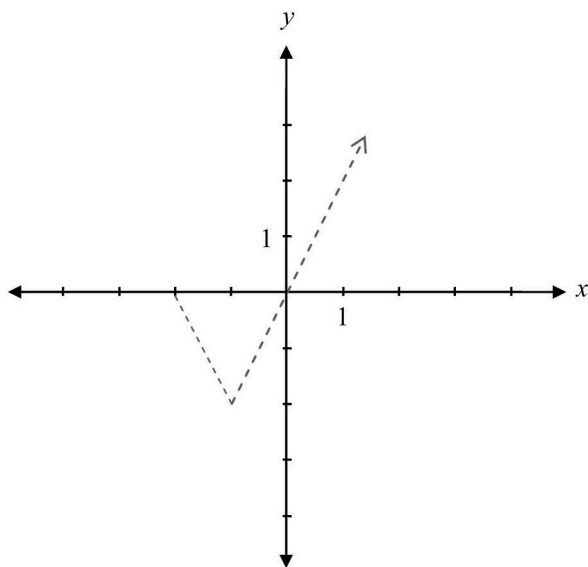
The value of $\sin \theta$ cannot be less than -1 .

Question 43

Given the graph of $y = f(x)$,



sketch the graph of $y = |f(2x)| + 1$.



The graph of
 $f(x)$ has
already been
drawn for your
reference.
No marks will
be awarded for
the graph of
 $f(x)$.

Solution

1 mark for absolute value 1 mark for horizontal compression 1 mark for vertical translation 3 marks y x 1 1 h h h y x

Question 44

Given $f(x) = 2^x + 1$, state the equation of the horizontal asymptote.

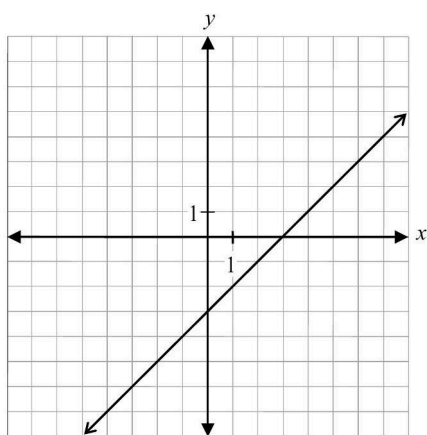
Solution

$$y = 1$$

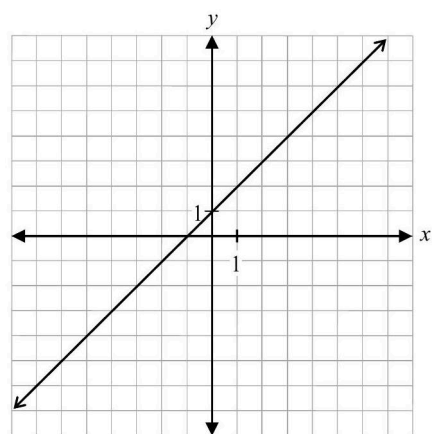
Question 45

Given the following graphs of $f(x) = x - 3$ and $g(x) = x + 1$,

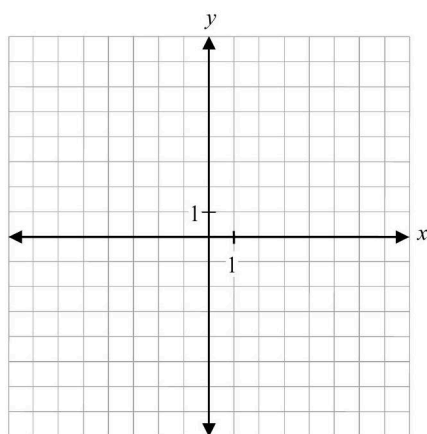
$f(x)$



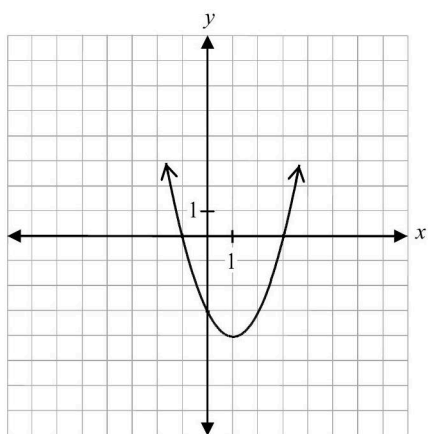
$g(x)$



sketch the graph of $h(x) = (f \cdot g)(x)$.



Solution



1 mark for operation of multiplication

1 mark for shape representing the given operation

2 marks